Possible Problems for Quiz 9 to be taken 3/28/19. Of these problems I will choose 3 to be on the quiz.

Problem 1

If the vectors $\vec{u}, \vec{v}$ have length 1 and $\vec{u} \cdot \vec{v} = \frac{\sqrt{3}}{2}$ what is the angle between the vectors?

Problem 2

How much of the vector $\vec{u} = \begin{bmatrix} 4 \\ 5 \\ 6 \end{bmatrix}$ points in the direction of $\vec{v} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$?

Problem 3

Let $\vec{u} = [1,2]\ and\ \vec{v} = [4,3]$ calculate $proj_{\vec{v}}(\vec{u})$ and use it to find a vector that is orthogonal to $\vec{v}$. Show it is orthogonal by calculating $\vec{u} \cdot \vec{v}$. Sketch $\vec{u}, \vec{v}, proj_{\vec{v}}(\vec{u})$ and the orthogonal vector on the same axes.

Problem 4

The set of vectors $\{\vec{v}_1 \ldots \vec{v}_k\}$ is orthogonal. Given this information find determine how many solutions there are to the following equation. Justify your answer.

$$c_1\vec{v}_1 + c_2\vec{v}_2 + \cdots c_k\vec{v}_k = \vec{b}$$

Problem 5

Write down an orthogonal basis for R^5.

Problem 6

Explain in a short paragraph why the $Gram - Schmidt$ process produces an orthogonal set of vectors.

Problem 7

Let $S = span\left\{\begin{bmatrix} -1 \\ 2 \\ 1 \end{bmatrix}\right\}$, find a basis for $S^{\perp}$.